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MA3521 — Variation of Parameters for Systems

Name:

1. **Trigonometric forcing.** Solve the initial value problem

$$\mathbf{y}' = A\mathbf{y} + \mathbf{f}(t) \qquad A = \begin{pmatrix} 0 & -3 \\ 3 & 0 \end{pmatrix} \qquad \mathbf{f}(t) = \begin{pmatrix} \cos(3t) \\ \sin(3t) \end{pmatrix} \qquad \mathbf{y}(0) = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

(a) Find the eigenvalues of A and an eigenvector for the eigenvalue with positive imaginary part. Use it to find two real homogeneous solutions.

(b) Use those solutions to form $\Phi(t)$ with $\Phi(0) = I$. Find $\det \Phi(t)$ and $\Phi(t)^{-1}$.

(c) Set $\mathbf{y} = \Phi\mathbf{u}$. Find $\mathbf{u}'(t) = \Phi(t)^{-1}\mathbf{f}(t)$ and integrate. Compare $\mathbf{f}(t)$ with the columns of $\Phi(t)$ before multiplying.

(d) Use the initial condition to find the constants and write $\mathbf{y}(t)$.

(e) Check the differential equation and the initial condition.

2. **Power forcing.** Find the general solution on $t > 0$.

$$\mathbf{y}' = A(t)\mathbf{y} + \mathbf{f}(t) \qquad A(t) = \begin{pmatrix} 0 & 1 \\ -12/t^2 & 6/t \end{pmatrix} \qquad \mathbf{f}(t) = \begin{pmatrix} 0 \\ t^3 \end{pmatrix}$$

The homogeneous system has the solutions

$$\mathbf{y}_1(t) = \begin{pmatrix} t^3 \\ 3t^2 \end{pmatrix} \qquad \mathbf{y}_2(t) = \begin{pmatrix} t^4 \\ 4t^3 \end{pmatrix}$$

(a) Verify that both vectors satisfy $\mathbf{y}' = A(t)\mathbf{y}$.

(b) Form $\Phi(t)$, compute its determinant, and find $\Phi(t)^{-1}$.

(c) Find $\mathbf{u}'(t) = \Phi(t)^{-1}\mathbf{f}(t)$ and integrate.

(d) Write $\mathbf{y}(t) = \Phi(t)\mathbf{u}(t)$ and identify the particular and complementary terms. Use c_1 and c_2 for the constants.

(e) Check the particular solution in the nonhomogeneous system.